

RolloPod – A Transformable Robot with Dual Mode Locomotion

Engineering Ideation Document

Executive Summary

Rollopod is a transformable robotic platform designed to bridge the gap between high-speed wheeled travel and high-adaptability legged locomotion. Traditional field robots are typically constrained by a fundamental compromise: wheeled platforms achieve high speeds on smooth surfaces but fail when encountering obstacles or fragmented terrain, while legged hexapods excel at stepping over rough obstacles but suffer from low travel speeds and high energy consumption on open paths.

Rollopod addresses this challenge through a reconfigurable mechanical architecture. Rather than carrying separate wheels, three articulated legs on each side of the robot fold inward edge-to-edge to form two continuous side rolling rings. In walking mode, all six legs extend to function as an articulated hexapod capable of navigating uneven terrain, climbing stairs, and crossing gaps. In rolling mode, the folded leg assemblies become the structural rolling wheels, enabling fast, energy-efficient differential travel.

A central suspended pod houses the primary control electronics, power systems, and perception sensors. By decoupling the central body from the rotation of the outer rolling rings using precision bearing interfaces and dynamic orientation balance, Rollopod maintains a stable, forward-facing sensor payload across walking, rolling, and transformation states. This document presents the design philosophy, mechanical kinematics, electronic architecture, control software framework, engineering challenges, and operational vision for the Rollopod platform.

Physical Architecture & Transformation Topology

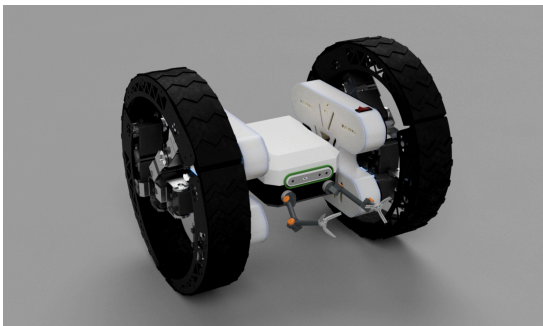


Fig.1: Rolling State

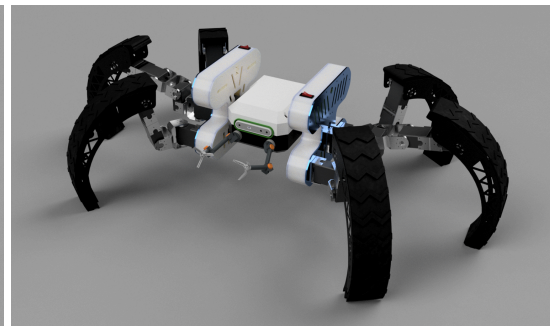


Fig.2: Walking State

Rollopod consists of three primary structural assemblies:

- 1) **Central Suspended Body Module:** Houses the primary electronic control hub, power distribution system, inertial sensing payload, and vision perception sensors. The central body remains suspended between the outer rolling structures across all operating states.
- 2) **Side Rolling Ring Assemblies:** Structural transformation frames mounted on either side of the central body via high-precision bearing interfaces.
- 3) **Transformable Articulated Leg Assemblies:** Six articulated legs in total (three mounted on the left ring assembly and three on the right ring assembly). During transformation, the three legs on each side fold inward to complete the circular outer rolling rings with treaded outer profiles.

Transformation States

Rollopod operates across three mechanically valid configurations:

- 4) **Walking State:** All six articulated legs are fully deployed outward, operating as a multi-DOF hexapod for obstacle climbing and uneven terrain traversal.
 - 5) **Rolling State:** Both side leg groups fold inward to complete the circular geometry of the side rolling rings, enabling high-speed differential rolling locomotion.
 - 6) **Transitional State:** Asymmetric or intermediate leg folding configurations maintained during state transitions, enabling dynamic mode switching while preserving body clearance and structural balance.
-

1. Design Philosophy & System Objectives

Mobility Trade-offs in Mobile Robotics

Mobile field robotics faces a fundamental trade-off between terrain adaptability and energy efficiency:

- 7) **Legged Locomotion:** Hexapods excel at negotiating complex, fragmented, and steep terrains by stepping over obstacles. However, articulated walking gaits incur high mechanical work and low linear travel speeds on flat ground due to continuous leg lifting and swinging cycles.
- 8) **Wheeled Locomotion:** Wheeled platforms offer high speed and low energy consumption on smooth surfaces, but struggle when encountering step obstacles, gaps, or steep inclines that exceed wheel radius.

The Rollopod Hybrid Philosophy

Rollopod resolves this trade-off by using a shared transformable mechanical structure rather than carrying separate, redundant wheel assemblies. The outer segments of the legs feature precision-curved outer profiles with heavy treads. When

folded inward, the leg links form two continuous structural rolling rings.

A primary design objective is **continuous**



environmental scanning. In traditional transformable or rolling robots, body rotation during rolling disorients perception sensors. Rollopod decouples central body orientation from ring rotation using central bearing assemblies and active balance control, ensuring that cameras, LiDAR, and sensors remain forward-facing during walking, rolling, and transformation.

2. Evolution of the Rollopod Concept

The development of Rollopod represents a progressive engineering journey spanning conceptual kinematics, CAD refinement, physical prototype iterations, and advanced control distribution:

- 9) Initial Conceptualization & Kinematic Vision:** The project originated from the goal of eliminating speed barriers in hexapod robotics without sacrificing terrain clearance. Early concepts explored spring-loaded mechanisms and passive return wheels for central body stabilization.
- 10) CAD Modeling & Mechanical Synthesis:** Detailed CAD models defined the linkage geometry, outer leg curvature, joint clearances, and ring folding kinematics. Simulation verified that three articulated legs per side could align edge-to-edge to form a circular rolling profile.
- 11) Prototype Iterations & Actuation Evolution:** Physical prototyping highlighted the limitations of centralized microcontrollers and passive mechanical balance. Passive spring balance was replaced by active differential motor control and central bearing pod isolation.
- 12) Active Distributed Architecture:** The control system evolved into a distributed architecture using a Master controller PC bridge, an on-robot Slave controller hub, multi-channel hardware PWM drivers, a dual-channel smart motor driver, and point-to-point wireless communication. High-level vision and spatial mapping were decoupled into an optional perception coprocessor architecture.
- 13) Future Vision – Modular Robotic Field Assistant (MRFA):** Rollopod's transformable dual-mode subsystem is designed to serve as the core mobile foundation for the **Modular Robotic Field Assistant (MRFA)**—a next-generation, autonomous field robotics architecture.
 - **Modular Payload Integration:** Future iterations of MRFA will feature standardized mechanical and electrical payload bays on the suspended central body. This allows rapid field swapping between thermal imaging units, gas detection sensors, robotic manipulator arms, and atmospheric measurement modules without altering core locomotion stability.
 - **Autonomous Field Missions:** Future MRFA platforms will execute multi-domain autonomous missions, including long-range environmental surveying, perimeter security, and hazardous area inspection. The system will autonomously determine when to roll for energy conservation and when to transition to legged walking based on real-time terrain mapping.
 - **Multi-Robot Swarm Capabilities:** Designed for multi-agent deployment, fleets of MRFA Rollopod units will coordinate wirelessly to perform collaborative search-and-rescue operations. Swarm units will share spatial maps, coordinate search grids, and establish relay communication networks across subterranean tunnels and collapsed structures.

- **AI-Driven Perception & Adaptive Locomotion:** Integrating onboard edge AI accelerators will enable real-time neural network terrain classification. The robot will predict terrain friction, slope angles, and structural integrity ahead of its path, dynamically adjusting leg stance width, ground contact force, or rolling velocity prior to entering challenging areas.
- **Disaster Response & Subterranean Exploration:** In post-disaster environments where human entrance is hazardous, MRFA units will penetrate rubble zones, stream real-time 3D spatial maps to remote incident command stations, and deploy emergency beacon payloads to assist emergency response teams.

3. Mechanical & Locomotion Architecture

3.1 Transformation Mechanism

Mode switching is driven by synchronized servo actuation across all six leg assemblies:

14) Unfolding Sequence (Rolling to Walking): The central controller commands joint servos to rotate outward, extending leg linkages from the ring structures to establish ground contact points.

15) Folding Sequence (Walking to Rolling): Joint servos draw the front, middle, and rear legs of each side inward until their treaded outer profiles align edge-to-edge, completing the circular rolling rings.



Fig 4.3.3: Transformation Steps

3.2 Walking Mechanism & Tripod Gait

In walking mode, Rollopod executes coordinated multi-leg gaits:

16) Tripod Gait Baseline: Legs are divided into two alternating triangular support tripods (Tripod A: Front Left, Rear Left, Middle Right; Tripod B: Front Right, Rear Right, Middle Left). While one tripod supports body mass, the opposite tripod advances.

17) Terrain Adaptation: Telemetry from an onboard IMU and stereoscopic 3D depth camera dynamically adjusts individual leg extension lengths and body posture over uneven ground.

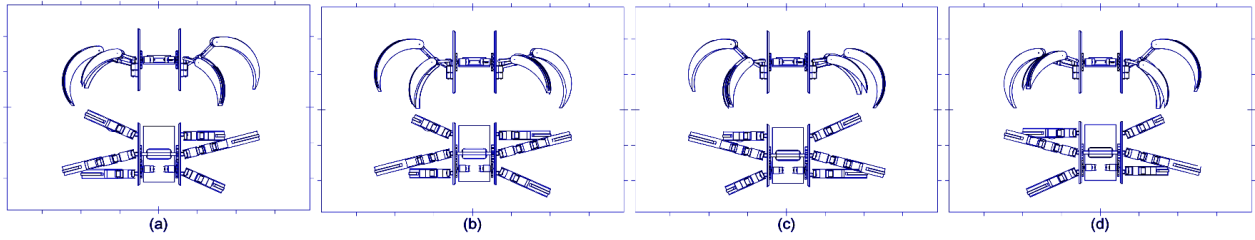


Fig 4.3.4: Forward Movement using Tripod gait pattern

3.3 Stair Climbing & Obstacle Traversal

For vertical step and stair negotiation, Rollopod executes a structured stance sequence:

1. **Approach & Low-Center Stance:** The robot lowers its center of gravity to maximize stability.
2. **First-Step Anchor:** The front leg pair extends upward to anchor securely onto the step surface.
3. **Body Elevation:** High-torque joint actuators elevate the central suspended body upward and forward onto the step.
4. **Sequential Ascent:** The middle and rear leg sets ascend in sequence to complete the step climb.

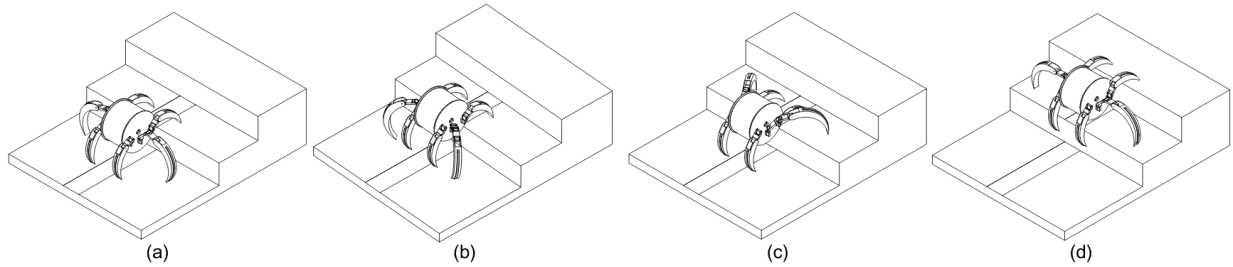


Fig 4.3.5: Climbing mechanism

3.4 Differential Rolling Mechanism & Body Isolation

Rollopod's rolling mode relies on active motor actuation and precision mechanical decoupling:

- 18) **Direct Differential Ring Drive:** High-torque DC motors drive each structural rolling ring independently through reduction gears powered by a dual-channel smart motor driver.
- 19) **Differential Steering:** Independent speed control of the left and right rolling rings enables forward travel, reverse motion, and zero-radius differential turning.
- 20) **Central Pod Bearing Isolation:** High-precision central ball bearing assemblies isolate the central body pod from outer ring rotation. This mechanical decoupling ensures that perception sensors remain stable and forward-facing during continuous rolling.
- 21) **IMU Orientation Indexing:** An onboard IMU monitors structural ring orientation. Prior to unfolding into walking mode, gyroscopic feedback indexes the rings so that legs deploy downward toward the ground, ensuring upright transformations.

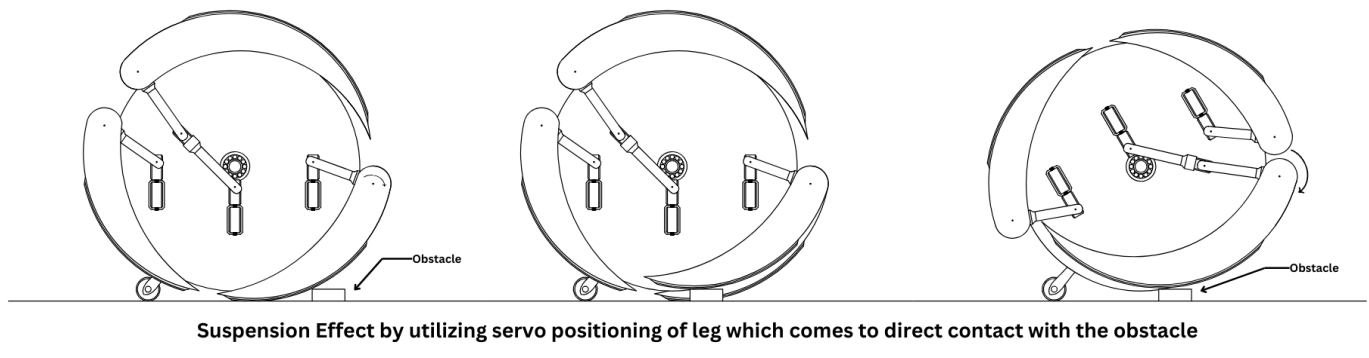


Fig 4.3.8: Motion Study for Rolling

4. Engineering Challenges & Design Trade-offs

Developing a transformable dual-mode robotic platform introduces complex engineering challenges across mechanical, electrical, and control domains. Addressing these challenges required deliberate architectural trade-offs:

4.1 Mechanical Transformation & Linkage Clearance

- **Challenge:** Aligning three articulated legs per side to form a smooth, circular rolling ring requires extremely tight mechanical tolerances. Any misalignment or clearance overlap prevents continuous rolling or causes joint binding during folding.
- **Design Trade-off:** Curved outer leg profiles with treaded contact surfaces were synthesized in CAD to provide edge-to-edge alignment when folded while maintaining structural stiffness during legged walking.

4.2 Mass Distribution & Center of Gravity Shifts

- **Challenge:** During transformation, leg mass moves dramatically relative to the central body. In rolling mode, shifting mass can induce unwanted body oscillation or cause the central pod to rotate with the rings.
- **Design Trade-off:** Heavier components (batteries, motor drivers, control hubs) are concentrated low within the central suspended pod, maximizing rotational inertia and keeping the center of gravity below the ring axis for natural pendulum stability.

4.3 Power Distribution & Inductive Load Isolation

- **Challenge:** High-torque servos and ring DC motors draw large, rapid current spikes during ground impacts and transformations. Sharing a single power bus causes voltage dips that reset sensitive control logic and corrupt digital sensor communication.
- **Design Trade-off:** Implemented a distributed dual-power architecture. Logic control electronics are powered by an isolated logic battery rail, while high-current actuators draw from separate actuator battery packs connected via a unified common ground bus.

4.4 Multi-Axis Actuator Synchronization

- **Challenge:** Transforming six multi-DOF legs simultaneously requires microsecond-level joint coordination. Asymmetric leg unfolding during motion can tip the robot over or lock leg links against the frame.
- **Design Trade-off:** Offloaded joint timing from the main processor to dedicated multi-channel hardware PWM drivers (such as dual PCA9685 controllers) driven by structured kinematic state machines.

4.5 Structural Rigidity vs. Mass Constraints

- **Challenge:** The chassis must withstand heavy dynamic ground impacts during walking while remaining lightweight enough for high-speed rolling and extended battery endurance.
- **Design Trade-off:** Combined high-strength aluminium extrusion structural frames with CNC-machined carbon fibre linkage arms to maximize stiffness-to-weight ratio.

4.6 Transformation Repeatability & Ground Indexing

- **Challenge:** In rolling mode, the side rings rotate continuously, meaning the legs could be oriented upside-down when transformation is requested.
- **Design Trade-off:** Integrated IMU gyroscopic attitude feedback on the rolling assemblies to index ring position, ensuring legs rotate downward toward the ground before opening into walking mode.

4.7 Distributed Control Latency & Real-Time Constraints

- **Challenge:** Teleoperating or autonomously controlling a dual-mode robot requires instantaneous command response for balance recovery and emergency stopping. Standard Wi-Fi network latency is unpredictable.
 - **Design Trade-off:** Adopted a dedicated point-to-point wireless protocol (ESP-NOW) between host and robot controllers, achieving low transmission latency (~10–30 ms).
-

5. Electronic System Architecture

Rollopod utilizes a distributed electronic architecture designed to decouple high-current actuator dynamics from control logic.

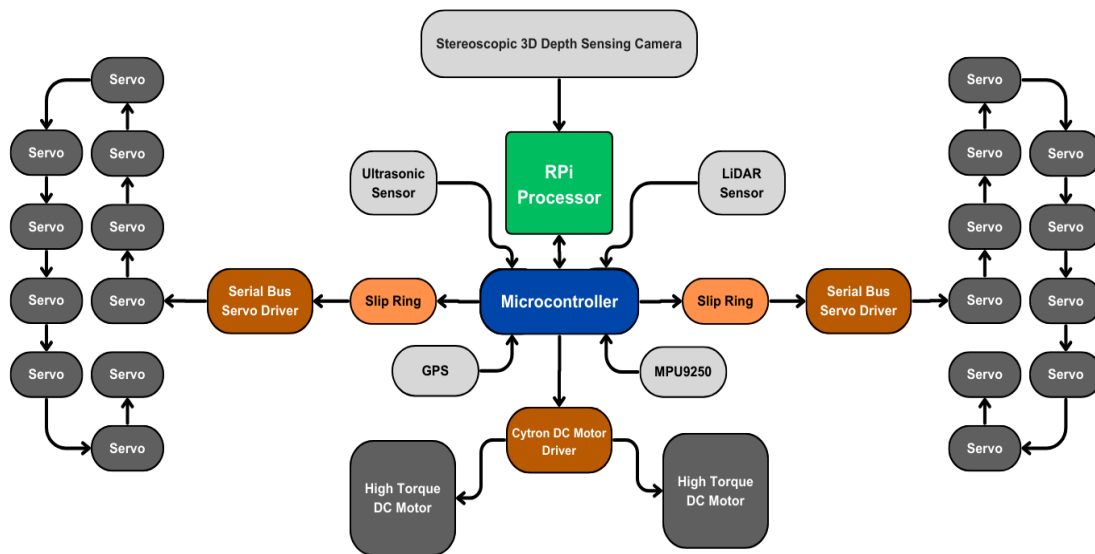


Fig 4.2.1: Electronic System Architecture

Core Electronics Subsystems

22) Master Control Hub (Host Wireless Bridge): A dedicated microcontroller (such as an ESP32 bridge) connected to the host control station. Translates operator inputs into low-latency, point-to-point wireless command packets (using ESP-NOW protocol).

23) Slave Control Hub (On-Robot Central Controller): An on-robot microcontroller (ESP32) mounted within the central suspended body. Serves as the central real-time controller, managing wireless command packets and distributing actuation orders across digital hardware buses:

- **Digital I2C Bus:** Communicates with multi-channel hardware PWM drivers and IMU sensors.
- **Motor Control Interfaces:** Sends directional logic and speed control signals to the ring motor driver.

24) Actuator Driver Subsystem:

- **Hardware PWM Drivers:** Multi-channel 12-bit PWM controllers (such as dual PCA9685 drivers) generate hardware-timed 50 Hz PWM signals for leg articulation and transformation servos, eliminating software timing jitter.
- **Smart Motor Driver:** A dual-channel high-current motor driver (such as a Cytron DC driver) provides independent differential speed and direction control to the rolling ring DC motors.

25) Sensing Subsystem:

- **Inertial Measurement Unit (IMU):** An MPU6050 6-DOF sensor provides dynamic pitch, roll, and angular velocity telemetry for stance balance, body level tracking, and transformation ring orientation indexing.
- **Spatial Perception Sensors:** An Intel RealSense 3D depth camera and LiDAR scanner capture environmental point clouds for obstacle classification and autonomous path planning.
- **Optional Perception Coprocessor:** High-level vision processing and spatial AI mapping can be offloaded to an optional onboard coprocessor (such as a

Raspberry Pi 5), keeping real-time joint control dedicated to the microcontroller network.

26) Distributed Power Architecture:

- **Logic Power Rail:** Dedicated LiPo battery supply (3S 5000mAh) powering control logic, sensors, and driver ICs through regulated buck conversion.
 - **Actuator Power Rail:** Independent high-current LiPo battery packs (3S 6200mAh/2500mAh) feeding servo power rails and motor driver power terminals, ensuring full electrical isolation.
 - **Unified Common Ground Topology:** Ties all power grounds and logic references together to maintain signal integrity across digital communication buses.
-

6. Software & Control System Architecture

Rollopod's software architecture is organized into modular functional layers:

Modular Software Layers

- 27) User Interface & Control Layer:** High-level control software providing teleoperation, joint range calibration, stance execution, and live telemetry feedback.
- 28) Wireless Communication Layer:** Low-latency point-to-point wireless transmission (ESP-NOW) ensuring fast packet delivery between host and robot controllers for real-time motion commands and emergency stops.
- 29) Motion Kinematics & State Controller:**
- **Gait Kinematics:** Computes inverse kinematics and coordinated joint trajectories for hexapod walking gaits (tripod gait baseline).
 - **Transformation State Machine:** Manages multi-axis joint routines for smooth transitions between walking, transitional, and rolling states.
 - **Differential Steering Kinematics:** Maps directional velocity commands into left and right rolling ring motor speeds.
- 30) Sensor Fusion & Spatial Perception:** Fuses IMU attitude telemetry with 3D depth sensing to maintain body levelness, measure terrain profiles, and navigate obstacles autonomously.
- 31) Safety & Emergency Management:** Monitors tilt limits, communication loss, and over-current conditions, triggering emergency crouch or motor stop routines to prevent structural damage.
-

7. Engineering Assembly Methodology

Assembly of Rollopod follows a structured modular engineering methodology:

32) Structural Framework & Bearing Integration:

- Construct the central suspended body pod using high-strength aluminium extrusions and CNC carbon fibre plates.
- Press-fit precision central ball bearing assemblies to mount the left and right structural rolling ring frames to the central pod.

33) Actuation & Drive Mechanism Mounting:

- Install high-torque joint servos onto CNC-machined mounting brackets on the leg linkage assemblies.
- Secure high-torque DC motor drive assemblies with reduction gearing onto the ring drive frames.

34) Distributed Electronics & Sensor Integration:

- Mount the central control hub, IMU, and depth camera centrally within the suspended body pod.
- Secure the hardware PWM drivers and motor drivers adjacent to their respective actuator banks to minimize signal noise.

35) Power System & Continuous Rotation Interface:

- Install logic and actuator battery packs, power switches, and common ground distribution.
- Install rotary slip rings across rotating ring interfaces to pass power and signal lines without wire binding during continuous rolling.

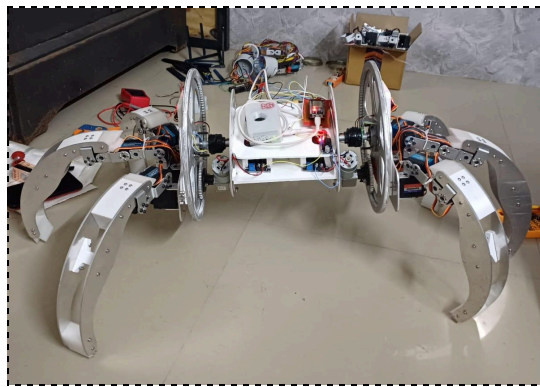


Fig 4.3.10: Assembly of Aluminium Plated Hexapod with Rolling Feature (Trials)

8. Testing & Engineering Validation Strategy

Verification of Rollopod follows a multi-stage engineering testing protocol:

36) Power Isolation & Bus Integrity Validation:

- Measure logic rail voltage stability under peak servo current loads to confirm brownout immunity.
- Validate common ground bus continuity across all battery sections and driver boards.

37) Wireless Communication & Latency Testing:

- Evaluate wireless packet delivery rates, link stability, and transmission latency across field operating ranges.
- Confirm digital I2C bus scanning and multi-device address responsiveness.

38) Joint Kinematics & Servo Range Calibration:

- Perform joint range calibration to establish precise mechanical zero positions and travel limits.
- Verify smooth multi-joint trajectory execution without mechanical interference.

39) Locomotion & Differential Drive Validation:

- Conduct walking tests to measure tripod gait speed, ground contact stability, and step clearance over obstacles.

- Evaluate motor driver differential rolling performance, measuring linear speed, acceleration, and zero-radius turning.

40) Transformation Repeatability & Dynamic Balance Testing:

- Test full transformation cycles between walking and rolling states to verify mechanical alignment and joint locking.

Component Category	Component Description / Model	Estimated Cost (INR)
Structural Components	Aluminium Extrusion Frame / Carbon Fibre Members	₹30,000
	CNC-Cut Aluminium Linkages & Structural Plates	₹6,000
	Actuator Mounting Brackets	₹2,000
	Precision Fasteners, Nuts & Bolts	₹3,000
	Precision Central Ball Bearing Assemblies	₹1,000
Motion Components	High-Torque Servo Motors (x18 core leg/transformation joints)	₹50,000
	High-Torque DC Motors (x2 side rolling ring drive)	₹7,000
Electronics & Power	Distributed 3S LiPo Battery Packs & 3S BMS Systems	₹8,000
	Intel RealSense 3D Depth Sensing Camera	₹39,000
	Raspberry Pi 5 (Optional High-Level Perception Coprocessor)	₹9,000
	Master & Slave ESP32 Microcontroller Boards (x2 Dev Modules)	₹1,500
	Dual PCA9685 16-Channel 12-Bit PWM Drivers (x2 Boards)	₹1,500
	Cytron Dual-Channel Smart DC Motor Driver Board	₹3,500
	LiDAR Sensor (Spatial Range Scanner)	₹13,000
	Rotary Slip Ring Connectors (Continuous Rotation)	₹9,000

	MPU6050 6-DOF IMU Sensor Board	₹1,000
	Power Switches, Schottky Protection Diodes & Buck Converters	₹1,500
Accessories	Wiring Assemblies, Connectors & Hardware	₹3,000
Total Estimated Cost	₹1,89,500	

Table 2.1: RolloPod Component Cost Analysis. Market research based on vendor catalogues including Robu, Robokits, RoboticDNA, and Amazon.

10. Assembly & Technical Illustrations

- **Figure 2.1:** Isometric View of Rollopod Assembly in Extended Hexapod Walking Mode.



Fig 2.1: Robot Assembly

- **Figure 2.2:** Intermediate Transformation Sequence Diagram.



Fig 2.2: Robot Transformation

- **Figure 9.1:** Rolling Motion Study Diagram.

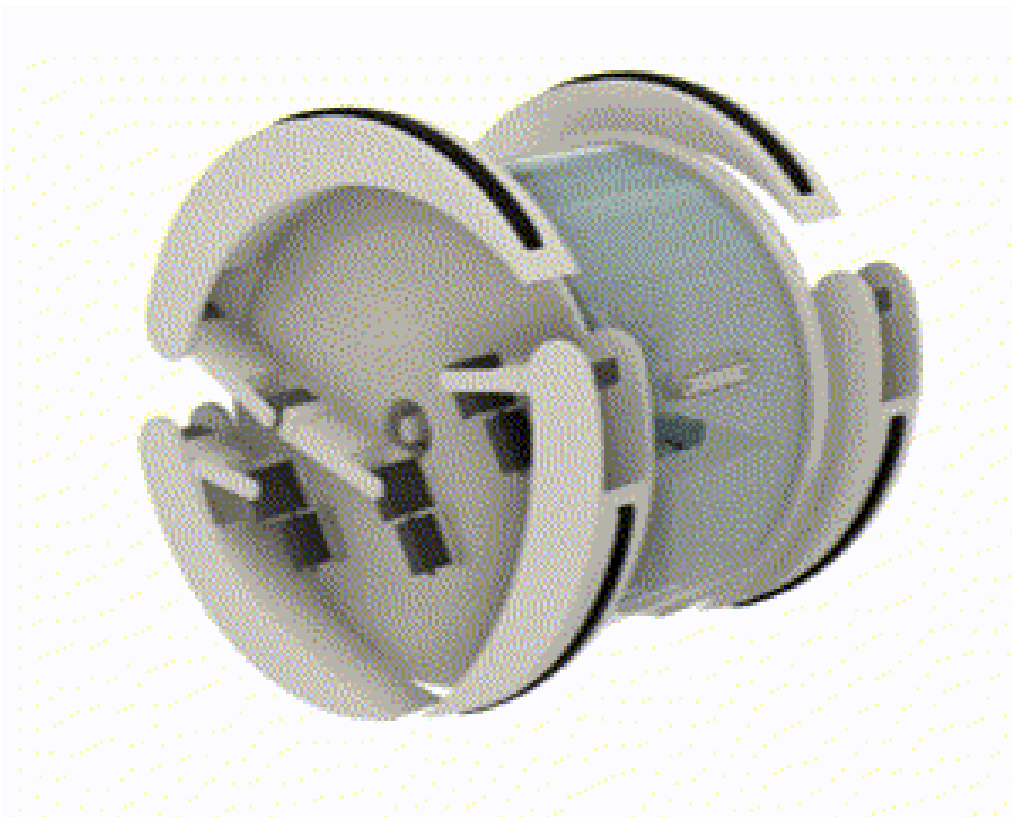


Fig 9.1: Rolling Motion study of hexapod gif

- **Figure 9.2:** Hexapod Transformation Diagram.

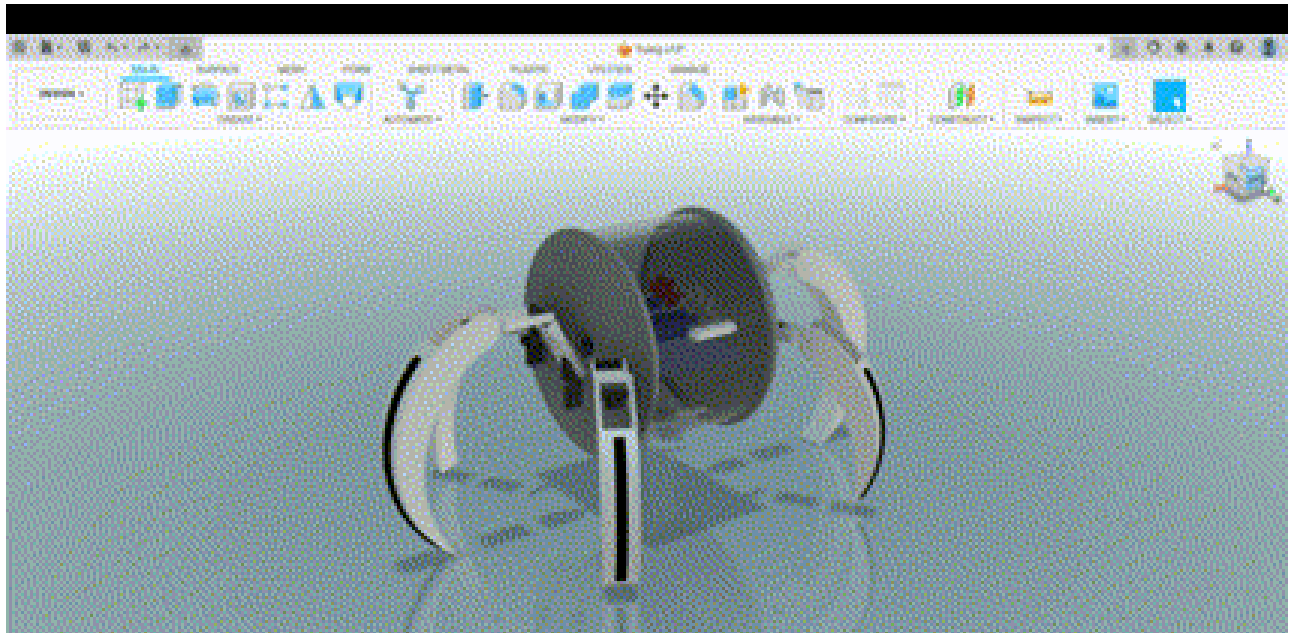


Fig 9.2: Hexapod Transformation gif

- **Figure 9.3:** Environmental Scanning via Depth Perception Payload.

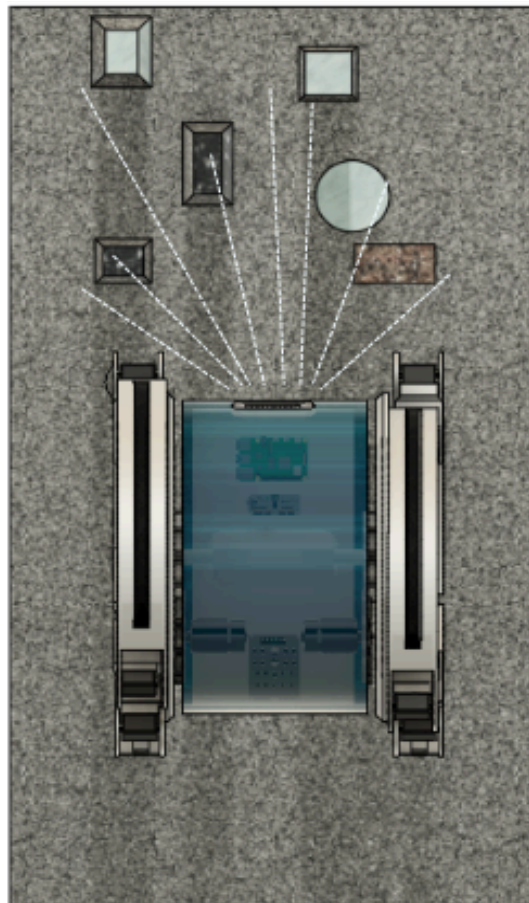


Fig 9.3: Environment Scanning via Depth Camera

- **Figure 9.4:** Rolling Locomotion Field Concept rendering.



Fig 9.2: Hexapod in Rolling Motion on Roads

- **Figure 9.5:** Agricultural Survey Field Concept rendering.



Fig 9.3: Hexapod in agricultural application

- **Figure 9.6:** Exploration Field Concept rendering.



Fig 9.4: Hexapod in exploration application

11. Operational & Societal Applications

Rollopod's hybrid dual-mode capability offers operational advantages across diverse field domains:

- **Search and Rescue Operations:** Rapidly travels over paved roads in high-speed rolling mode, transforming into legged walking mode to navigate collapsed rubble, disaster debris, and narrow openings while maintaining stable sensor orientation for survivor detection.
 - **Industrial & Infrastructure Inspection:** Inspects pipeline corridors, underground culverts, and industrial facilities, switching between efficient rolling on flat surfaces and legged stepping over curbs, steps, and conduits.
 - **Agricultural Automation:** Traverses crop fields for soil sampling, crop health monitoring, and spot treatment without compacting soil or damaging plant rows.
 - **Exploration & Planetary Robotics:** Provides a survey platform for caves, deserts, and unmapped planetary terrain, combining long-range rolling efficiency with legged obstacle traversal.
 - **Defense & Tactical Reconnaissance:** Conducts perimeter surveillance, hazard assessment, and logistics support in unpredictable terrain.
 - **Education & Robotics Research:** Serves as an open, modular research platform for bio-inspired locomotion, hybrid dynamic transformations, and distributed control systems.
-

12. Physical Specifications & System Dimensions

- **Overall Length:** 41.5 cm
 - **Overall Width:** 50.4 cm
 - **Overall Height:** 41.5 cm
 - **Primary Frame Construction:** High-Grade Aluminium Extrusion & Mild Steel for Legs
 - **Locomotion Modes:** 6-Leg Hexapod Walking / Dual-Ring Differential Rolling
-

13. Comprehensive Conclusion & Future Outlook

Rollopod presents a transformable robotic architecture that resolves the mobility trade-off between speed and terrain adaptability. By synthesizing a six-legged hexapod and a dual-ring rolling robot into a single reconfigurable mechanical structure, Rollopod demonstrates that a mobile platform can achieve high-speed continuous travel without sacrificing the ability to step over obstacles, climb stairs, or navigate fragmented field environments.

Core Engineering Contributions

- 41) Shared Structural Transformation:** Rather than carrying passive permanent wheels, Rollopod utilizes the articulated legs themselves as the structural rolling rings. Folding three curved, treaded leg assemblies per side edge-to-edge creates two functional rolling wheels, minimizing dead weight and maximizing mechanical efficiency.
- 42) Payload & Sensor Decoupling:** Central precision ball bearing assemblies and dynamic orientation control isolate the suspended body from outer ring rotation. This ensures that perception sensors, vision payloads, and control hubs maintain a horizontal, forward-facing orientation during continuous rolling, walking, and state transformations.
- 43) Distributed Control Architecture:** Moving from centralized microcontrollers to a distributed hardware network (incorporating Master/Slave wireless microcontrollers, multi-channel hardware PWM drivers, smart motor drivers, and isolated power rails) eliminates actuation jitter, prevents logic brownouts, and delivers deterministic multi-axis control.
- 44) Field-Proven Design Methodology:** Integrating CAD kinematic modeling, multi-phase engineering validation, and modular physical prototyping establishes a repeatable methodology for transformable field robotics.

Long-Term Potential & Technological Impact

Rollopod represents more than a single prototype; it establishes a scalable foundation for hybrid field robotics. As mobile robots are increasingly deployed in unpredictable, hazardous, and unmapped environments, the ability to dynamically adapt locomotion modes will become essential.

The Rollopod architecture lays the groundwork for the **Modular Robotic Field Assistant (MRFA)**, where field units equipped with AI perception, modular payload bays, and swarm communication networks will execute autonomous search, rescue, environmental monitoring, and infrastructure inspection missions. By proving that mechanical reconfigurability and

continuous environmental perception can coexist in a lightweight platform, Rollopod advances the field of adaptive robotics toward safer, faster, and more versatile autonomous operations.

Design Authorship

Respectfully Submitted by:

- Parag Patil
- Rutu Patel

----- Thank You -----